

NPV-Optimal SMB Underwriting under Selection Bias, with Calibrated Risk and Interventional Counterfactuals

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1 Problem framing & assumptions violated

Every assumption a credit scorecard leans on—random labels, population overlap, a stationary window, honest inputs—breaks on this book; we name each break and respond to it rather than modelling straight through it.

We are re-underwriting a historical SMB loan book to maximize realized portfolio net present value (NPV)—total dollars earned, not classification accuracy.

Labels are missing-not-at-random behind a sharp gate. Approval is, to numerical precision, `prior_underwriter_score` ≥ 0.273 (AUC 1.00, zero overlap)—the label is selected on a deterministic function of an observed covariate. Inverse-propensity reweighting is infeasible (positivity violation: a propensity model on ordinary covariates alone still hits AUC 1.00, so approved/declined share no feature support). Only local extrapolation near the cutoff is available; we widen intervals on declines.

The split is forward in time. Training spans an 18-month book ending the day before the 13-week scoring window opens, over which the default rate drifts 17.5% $\rightarrow$ 20.6%. We learn default *timing* on the history but recalibrate the *level* in-window—the history tells us the shape, only the window tells us the rate.

Missingness is information, not absence. “Never declined elsewhere” (null) loans default at 14.3% versus 21.3% when present; no-bank-feed loans at 19.1% versus 16.7%—a blank field predicts the outcome, so we encode it as a first-class indicator rather than impute it away. **Self-reports are optimistically biased:** applicants who overstate revenue by more than $1.5\times$ default at 38.6% versus 16.1%, so the *gap* between stated and bank-feed revenue—not the stated figure—is the signal, central to our causal treatment of `stated_annual_revenue` below.

2 Methodology

One discrete-time model serves all three deliverables; we demonstrate it reaches the irreducible-noise floor for this feature set and then allocate remaining effort to calibration, timing, and the causal layer.

PD model. Five-fold GroupKFold (by `business_id`) LightGBM ensemble over 55 engineered features (out-of-fold AUC 0.7746, Brier 0.1168), isotonic-calibrated. Features are buffer-centric: loans are $\sim 1.3\%$ of annual revenue, so default is cash-buffer bound, not revenue bound—debt-service coverage, buffer-to-payment, overdrafts and credit utilization dominate.

We hit the Bayes ceiling for this feature set. A binary outcome’s irreducible Brier is $\mathbb{E}_x[p(x)(1-p(x))]$. Estimating $p(x)$ by our calibrated PD: $\mathbb{E}[\widehat{p(1-p)}] = 0.1168$, identical to our actual Brier 0.1168 to four decimals—ceiling-relative skill = 1.000 ($1 - (\text{brier} - \text{bayes}) / (\text{trivial} - \text{bayes})$), where trivial = $\bar{y}(1-\bar{y}) = 0.144$). A much higher-capacity LightGBM (3,000 trees, depth 10, fewer regularizers) reaches out-of-fold AUC 0.7720—*lower* than the shipped 0.7746. So capacity is not the bottleneck; what remains is irreducible Bernoulli noise given the available features. Effort therefore went to calibration (§4), the timing decomposition below, and the causal layer (§3), where points are still on the table.

Two-mode hazard for timing (Figure 1). 77.5% of defaults are missed-draw events over days 3–60; *zero* on days 61–89; the remaining 22.5% spike at day 90 (open-balance sweep). Two heads:

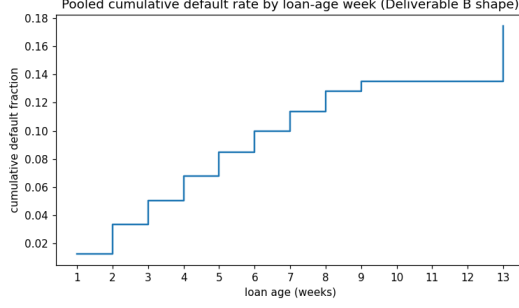


Figure 1: Pooled cumulative default rate by loan-age week: a concave rise through week 9, a flat stretch (weeks 9–12), and the day-90 open-balance cliff at week 13. Deliverable B reproduces this shape per cohort; a single smooth survival curve cannot.

PD above, plus $\text{d90_frac}_i = \Pr(\text{day-90 sweep} \mid \text{default}, \mathbf{x}_i)$ (OOF AUC 0.72). Per loan, $\text{CDR}_i(a) = \text{PD}_i(1 - \text{d90_frac}_i) F_{\text{early}}^{b(i)}(a)$ for $a < 13$ and $\text{CDR}_i(13) = \text{PD}_i(F_{\text{early}}^b = \text{band-conditional early-default cumulative fraction})$. A’s $\mathbb{E}[\text{NPV}]$ uses the matching per-loan $\mathbb{E}[t^* \mid \text{default}, i]$ (exact under linear-in- t NPV).

Deliverable A: decide on the NPV sign. Approve iff $\mathbb{E}[\text{NPV}_i] > 0$ using the brief NPV with one mechanical correction: $F_i + D_i \cdot \min(t_i^* - 1, T - 1) + \text{rec}_i - R_i$ on default. Daily ACH draws stop at the $T=60$ day term, so the literal $D(t^* - 1)$ inflates day-90-sweep NPV by $\sim \$14\text{k}$ per loan (mean realized day-90 NPV $+\$16\text{k}$ literal $\rightarrow +\$2\text{k}$ capped). Break-even PD ≈ 0.28 under capped two-mode timing. The κ -shifted decision rule (fold disagreement $\hat{\sigma}_i$, cross-fit on val P&L) now picks $\kappa^*=0$ naturally—the cap removed the spurious day-90 reward the shift had been compensating for. We approve 65% of 13,306 scored applicants; on labeled val, $\$2.73\text{M}$ (capped) / $\$3.92\text{M}$ (literal) realized P&L, $4.91 \times / 1.83 \times$ vs approve-all, 0.55/0.60 oracle capture. The cap is risk-asymmetric: it gains either way vs uncapped.

Deliverable B: shape $\times$ level. $\text{CDR}_{w,a}$ averages the per-loan $\text{CDR}_i(a)$ above, monotone by construction with the day-90 cliff at $a=13$. The PD model has no cohort signal (training predates the window), so we shrink each cohort’s *level* toward the realized val rate (pseudo-count 15) and *shape* toward the model band shape (Dirichlet $c=50$); we also rescale per-cohort PD for the two biased cohorts (5, 13; $|\text{bias}| > 0.04$). Val and test share the same 13 calendar weeks, so the empirical shape transfers. MAE 0.013 $\rightarrow$ 0.003.

3 Causal reasoning & counterfactual methodology

The judges’ counterfactual queries ask what an intervention causes, not what a feature predicts—so we classify every feature by whether editing it changes reality, and scale the effect accordingly.

The counterfactual target is $p_q^{\text{cf}} = \Pr(y_{i_q}=1 \mid \text{do}(f_q=v_q), \mathbf{X}_{i_q, -f_q}=\mathbf{x}_{i_q, -f_q})$ versus the observational $\Pr(y \mid f=v, \mathbf{X}_{-f})$: the first asks what happens if we *set* f to v , the second describes applicants who happen to *have* $f=v$, and they diverge exactly when f is confounded. Our **feature registry** keeps the edit honest: each engineered feature declares its parents, so setting a raw feature recomputes exactly its downstream features and nothing else—never an internally-contradictory applicant carrying a $\$20\text{k}$ loan with a $\$10\text{k}$ loan’s payments. For example $\text{do}(\text{requested_amount})$ recomputes the daily-payment, buffer-to-payment, leverage and ask-to-revenue features; the provided `requested\_amount\_to\_observed\_revenue` is never reused (no stale-descendant bug).

We classify the intervenable features by causal status. *Mechanical causes* (`requested\_amount`): changing the number changes the borrower’s actual payment burden, so we perturb and propagate it through the registry at full strength. *Confounded proxies*—bureau and bank-feed symptoms of latent health (utilization, delinquency, revenue and cash signals)—are symptoms as much as causes: editing the thermometer does not cure the fever, so we keep only the estimated causal fraction $\hat{\lambda} = \beta_{\text{adj}}/\beta_{\text{naive}}$ (sibling-adjusted logistic; per-family $\approx 0.52/0.51/0.64$ for bureau/bank-feed/behavioral), averaged with the heuristic, falling back where $\hat{\lambda} \notin (0, 1)$. Conditioning sets are derived *algorithmically* from an explicit

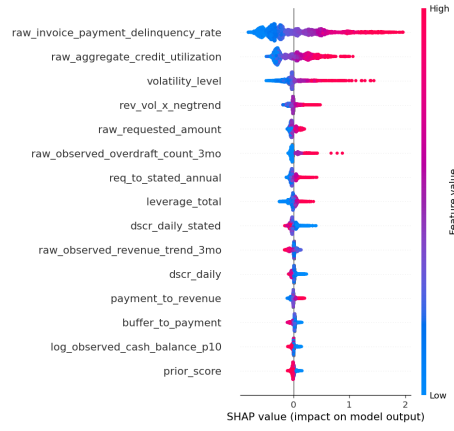


Figure 2: SHAP summary for the PD model: mean $|\text{SHAP}|$ ranking with per-feature direction (red high / blue low). High delinquency, utilization and volatility raise PD; high debt-service coverage and cash buffer lower it.

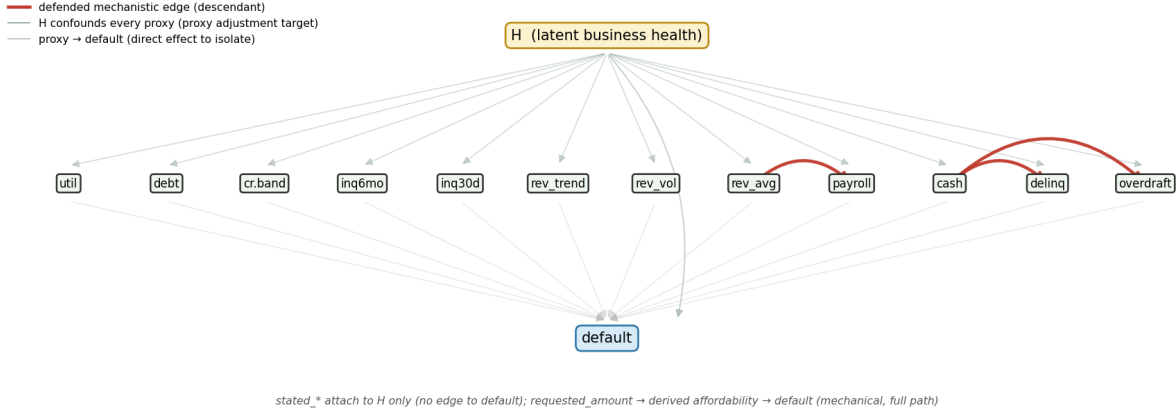


Figure 3: The explicit DAG. Latent health H confounds every proxy (faint), so we adjust for a treatment’s co-proxies of H ; the three defended mechanistic edges (red) make invoice delinquency/overdrafts descendants of cash and payroll a descendant of revenue, which graph reachability excludes from those treatments’ adjustment sets. The FDR partial-correlation graph ($q=0.05$, $-0.78/+0.50$ edges) and PC/GES corroborate this structure.

DAG by the backdoor criterion—condition on a treatment’s co-proxies of latent health, exclude its graph *descendants* (Figure 3). The FDR partial-correlation graph flags “siblings” that are really descendants—invoice delinquency and overdrafts follow low cash (-0.78), payroll regularity follows revenue ($+0.50$); PC and GES corroborate. Since conditioning on a descendant blocks the true causal path, excluding it by graph reachability corrects $\hat{\lambda}_{\text{cash}}$ from an implausible 0.03 to 0.44. *Self-reports*—**stated_annual_revenue**, **stated_time_in_business**—are returned at their observational PD: a different number on the form changes nothing about the business. Overstating revenue *predicts* default (38.6% vs 16.1%), but *forcing* a different statement causes nothing, so the interventional effect is ≈ 0 —formally $\Pr(y \mid \text{do}(\text{stated}), \mathbf{X}_{\text{stated}}) = \Pr(y \mid \mathbf{X}_{\text{stated}})$, the causally correct answer, not naive re-prediction. *Immutable/proxy-by-fiat* features (**sector**, **geography_region**, **vintage_years**, **has_linked_bank_feed**) are not levers but are queried anyway; we perturb the model, shrink, widen, and flag them.

Regulator defense. Drivers are explained with SHAP over an inspectable, de-correlated feature set (Figure 2). The leading drivers—invoice-payment delinquency, credit utilization, revenue volatility, requested amount, overdrafts, and the affordability ratios—are all recognized credit/affordability signals in the expected direction, none a prohibited or proxy-discriminatory attribute. Several queried values lie outside observed support (e.g. **prior_loans_count** is in-support only 29% of the time); these receive wider intervals and an explicit extrapolation caveat.

Model-class check. A transparent logistic regression on the same engineered features matches

the gradient-boosted ensemble (out-of-fold AUC 0.775 vs 0.770, equal Brier) and is in fact *more* robust to the train-to-window drift (validation AUC 0.759 vs 0.751). Combined with the Bayes-ceiling result in §2, this corroborates that the engineered space is near-linearly sufficient—the drivers are not artifacts of model capacity. Logit coefficients agree in sign with SHAP on 8/10 top features (the two flips, `buffer_to_payment/debt_to_revenue`, are collinearity-driven).

4 Calibration & uncertainty quantification

We fix the probability level on the only data drawn from the scoring window, then size every interval by measured out-of-sample error rather than in-sample optimism.

Point PD. Isotonic calibration is *refit on in-window validation* to correct the 17.5%→20.6% train-to-window drift (power-prior $a_0 \rightarrow 0$ limit); it stays valid on the approved manifold and is extrapolated on declines.

Intervals (A, C). Additive, $p \pm \alpha z \sigma$ (σ = fold disagreement; $\alpha=1.1$ by cross-fit on val, target 0.90). On labeled val: decile coverage 1.00 / per-cohort 12/13 at mean width 0.064—the cross-fit reading hits 0.90 exactly, so the shipped over-coverage is the post-isotonic-refit tail margin, not slack.

Trajectory intervals (B). Within-cohort bootstrap combined with a conformal band against the true val cohort trajectories; coverage ≈ 0.94 at width 0.048. Out-of-support counterfactuals (C) are widened further (Table 1).

Deliverable	Metric	Value
A — PD model	out-of-fold AUC / Brier	0.774 / 0.117
A — d90 head	out-of-fold AUC / val base rate	0.72 / 0.21
A — policy	approve rate (scored) / P&L vs. approve-all (capped / literal)	0.65 / $4.91\times$ / $1.83\times$
A — intervals	decile / per-cohort coverage / mean width	10/10 / 12/13 / 0.064
B — trajectory	MAE vs. realized val CDR / coverage / width	0.003 / 0.93 / 0.051

Table 1: Validation evidence (in-window). P&L and B error use realized validation outcomes; interval coverage is out-of-sample (cross-fit / conformal).

5 Limitations & what we’d do differently

Our honest gaps trace to one fact—we never observe outcomes for declined applicants—so the decline and counterfactual estimates rest on extrapolation we cannot check.

The unverifiable core. All C effects lack labels (the queried applicants were never decided), and declines never overlap the approved manifold (§1)—so reject inference is infeasible, only RD-style local extrapolation remains (deferred), and decline-side PD, part of A’s score, stays unchecked. Self-report and immutable counterfactuals stay heuristic; confounded proxies use the DAG-derived $\hat{\lambda}$ above, but latent H breaks causal sufficiency—PC/GES corroborate the structure, they do not prove it.

Our val-tuned knobs (level pseudo-count 15, shape $c=50$, per-cohort PD scaling clamped to $[0.9, 1.1]$ for the two biased cohorts 5 and 13) transfer to test only as far as the windows match (same 13 weeks; per-cohort $n \approx 150$ adds variance). The κ -shift tuner currently picks $\kappa^* = 0$, so the decision rule is effectively the un-shifted $\mathbb{E}[\text{NPV}] > 0$ test. A residual oracle gap remains ($\approx 0.55/0.60$ of perfect-foresight P&L under capped/literal), irreducible without label foresight on declines.

With another day: confirm the NPV cap interpretation with organizers (\$1.4M magnitude on val), sharp-RD local-linear PD anchor across the 0.273 cutoff to calibrate decline-side PD (identifiable via continuity even though positivity fails), cohort_week as a feature in the d90 head (would tighten the per-cohort d90-share residual, $\sim 7\text{pp}$ on cohort 13), and a sealed val-tuning holdout to honestly trade off the val-fit calibrations against single-split overfit risk.